

BIO 110: Cell Biology

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Lecture 5. Membrane transport in the eukaryotic cell.

Prerequisites: phospholipid structure, fluid mosaic model.

The plasma membrane as a barrier

The plasma membrane is a phospholipid bilayer roughly 7 nm thick.

Hydrophobic interiors make the bilayer semipermeable: small non-polar molecules cross freely, ions do not.

Integral proteins span the bilayer and provide selective channels.

Passive transport

Simple diffusion moves a solute down its concentration gradient and consumes no ATP.

Facilitated diffusion moves a solute down its gradient through a channel or carrier protein.

Osmosis is the movement of water across a semipermeable membrane from low solute concentration to high solute concentration.

In a red blood cell placed in distilled water, water enters by osmosis and the cell lyses (haemolysis).

Active transport and the sodium-potassium pump

Active transport moves a solute against its gradient and requires energy.

The sodium-potassium pump exports three sodium ions and imports two potassium ions per ATP hydrolysed, so the pump is electrogenic.

Inhibition by ouabain stops the pump and the cell swells.



Summary table

Simple diffusion: down gradient, no protein, no ATP.

Facilitated diffusion: down gradient, protein, no ATP.

Primary active transport: against gradient, protein, ATP.

Endocytosis and exocytosis: bulk transport of vesicles, ATP required.

Tutorial question: explain why a plant cell in distilled water does not lyse as readily as an animal cell.